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Journal of Non-Crystalline Solids 292 (2001) 25–43

JOURNAL OF NON-CRYSTALLINE SOLIDS

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The effect of experimental conditions and evaluation techniques on the alteration of low activity glasses by vapor hydration

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Received 7 May 2001

Abstract

The vapor hydration test (VHT) is used to study the reaction between waste forms and water with the aim to gain insight into their alteration behavior. In nuclear-waste immobilization, the test is primarily used as a screening tool to identify durable/non-durable glasses and as a convenient method to generate and identify alteration products for use in performance modeling. The lack of a standard procedure for conducting this test has resulted in a number of reported test methods, thus decreasing the ability to directly compare test results from different sources. To optimize the VHT procedure, a series of tests was conducted on simulated low activity waste (LAW) glasses at temperatures ranging from 150 to 300 °C with different volumes of water, specimen holders, specimen-preparation techniques, and data-evaluation methods. Reaction progress was monitored by measuring the thickness of the remaining glass and the alteration layer. The alteration rate at which glass is converted into the alteration products was determined by linear regression from the remaining glass thickness plotted as a function of time. The resulting procedure for conducting VHT eliminates problems associated with the measurement of alteration layers and enables direct comparison of alteration rates for different materials. © 2001 Published by Elsevier Science B.V.

PACS: 28.41.K

1. Introduction

In recent years, a significant effort has been applied towards understanding the mechanism of hydration and alteration of glasses in aqueous media. One reason to study these processes is the humid repository environment, where hydration of glass surfaces may be an important aging mechanism for vitrified waste forms [1,2], subsequently influencing the release of radioactive and hazardous constituents. It is known, and has been demonstrated [3], that the precipitation of alteration products significantly influences the alteration process. Therefore, to understand the mechanism and related processes that occur during the reaction with humid air or an aqueous environment, it is necessary to generate and characterize the alteration products.

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PII: S0022-3093(01)00875-4

A test is needed in which the alteration process would be accelerated, and the identification of solid alteration products would be facilitated. However, it is necessary to demonstrate that the test accelerates the alteration process and generates alteration products similar to those that would be created by natural aging. Luo et al. [4] compared alteration products from naturally altered Hawaiian basaltic glasses to those from the vapor hydration test (VHT), conducted at temperatures ranging from 70 to 240 °C, and concluded that the VHT accelerates the generation of alteration products and results in formation of the same alteration products for the range of tested temperatures.

Generally, VHT is a test that can be used to (1) determine if a glass is likely to corrode at an extreme rate, (2) obtain alteration products within a short period, and (3) measure the rate at which glass is converted into alteration products at elevated temperatures. However, the lack of a standard procedure for conducting VHT decreases the ability to directly compare test results. Therefore, it is necessary to determine the effect of experimental conditions on test results. Similar studies have been performed for the product consistency test (PCT) [5,6], the Materials Characterization Center Static Leach Test 1 (MCC-1) [7], and the single-pass flow-through test (SPFT) [8,9] during the process of their standardization. This study presents a systematic evaluation of specimen preparation, mass of water used to conduct the test, and specimen evaluation method effects on the VHT response for several simulated LAW glasses. A procedure for conducting VHT [10] was developed based on the results discussed in this paper.

2. Experimental

A series of tests was performed on simulated LAW glasses to optimize the test parameters. The aim of these tests was to determine the effects of specimen preparation, volume of added water, and evaluation techniques on results and their reproducibility. Tests were conducted at temperatures ranging from 150 to 300 °C on LD6-5412, LAW-A33, LAW-ABP1, HLP-47, LAW-A44, HLP-01, HLP-25, HLP-26, and HLP-43 glasses. Their compositions are given in Table 1. Results for LD6-5412 glass from PCT, MCC-1, and VHT were reported by Feng et al. [12]. Results for LAW-A33 and LAW-ABP1 from PCT, VHT, and pressurized unsaturated flow test (PUF) were reported by McGrail et al. [13]. Results for HLP-47, LAW-A44, and HLP-01 from PCT and VHT were reported by Vienna et al. [14].

Glasses were prepared from single metal oxide and carbonate precursors for all components except SO_3, P_2O_5, F^-, and Cl^-, which were added as sodium salts, and B_2O_3, which was added as H_3BO_3. Batches sufficient to produce 500 g of glass were blended in an agate mill, melted in a covered Pt/Rh crucible for 1 h, quenched on a steel plate, ground in a tungsten carbide mill, and remelted in a covered Pt/Rh crucible for 1 h. Glass compositions were verified by chemical analyses [14]. Five glasses, HLP-01, LAW-A33, LAW-A44, HLP-47, and LAW-ABP1, were heat treated according to the cooling schedule, given by Eq. (1), directly after the second melt:

$$T\ (^\circ\mathrm{C}) = 925.7 - 20.9t\ (\mathrm{h}). \tag{1}$$

Glass LD6-5412 was not heat treated after the second melt, but poured into a $2 \times 2 \times 10\text{-cm}^3$ bar, annealed at 500 °C for 2 h, and allowed to cool slowly to the room temperature.

2.1. Description of VHT procedure

The apparatus for conducting VHT is displayed in Fig. 1. Specimens with dimensions $10 \times 10 \times 1.5\ \mathrm{mm}^3$ were prepared from glass bars with a diamond impregnated saw. Except specimens used for tests with different surface finish, all sides were polished to 600-grit surface finish with water-lubricated SiC paper. Specimens, stainless steel vessels, lids, and supports were cleaned, and specimens were held with Pt wire connected to a stainless steel support. The support with the specimen was placed inside the vessel together with a predetermined volume of deionized water. The sealed vessel was held at constant temperature in a convection oven for a preset time. After test termination, the specimen

Table 1
Composition of tested glasses (mass%)

Glass ID	SiO_2	Na_2O	Al_2O_3	B_2O_3	Fe_2O_3	TiO_2	ZnO	ZrO_2	MgO	K_2O	CaO	Cl^-	SO_3	P_2O_5	Cr_2O_3	F^-	La_2O_3
LD6-5412	55.91	20.41	12.17	5.05	0.11	0.02	0.14	0.08	0.00	1.66	4.1 2	0.00	0.18	0.22	0.05	0.00	0.00
LAW-A33[a]	38.25	20.00	11.97	8.85	5.77	2.49	4.27	2.49	1.99	3.10	0.00	0.58	0.10	0.08	0.02	0.04	0.00
LAW-A44	44.55	20.00	6.20	8.90	6.98	1.99	2.97	2.99	1.99	0.50	1.99	0.65	0.10	0.03	0.02	0.01	–
LAW-ABP1	41.89	20.00	10.00	9.25	2.50	2.49	2.60	5.25	1.00	2.20	0.00	0.58	0.10	0.08	0.02	0.04	2.00
HLP-47	54.71	20.12	10.06	8.05	1.01	0.10	–	1.01	0.10	1.51	0.50	0.80	0.20	0.50	0.20	1.01	–
HLP-01[b]	49.07	20.00	7.00	10.00	5.50	3.00	1.50	1.50	1.50	0.41	0.01	0.28	0.07	0.06	0.08	0.01	0.00

[a] LAW-A33 and LAW-A44 are glasses developed by the Vitrious State Laboratory at Catholic University of America for immobilization of Hanford low activity waste [11].
[b] HLP-01 contains also 0.01 mass% of ReO_2, and its target composition is identical to HLP-25, HLP-26, and HLP-43.

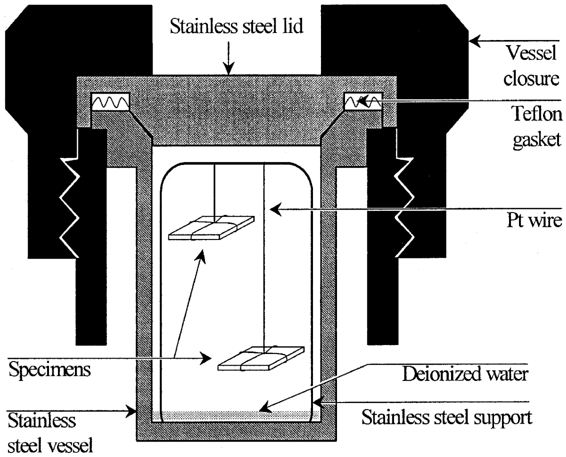

Fig. 1. Apparatus for conducting VHT.

was removed from the vessel and examined for the presence of alteration products, first by visual observation and then with optical microscopy (OM). The pH value of the remaining fluid in the vessel was measured with pH paper, which served as an indication of possible water reflux (see below). The thickness of the remaining, visibly unaltered glass and the layers of alteration products were measured on specimen cross-sections with image analysis (IA).

2.2. Equipment calibration

The initial dimensions of the specimens were measured with calipers. The thickness of unaltered glass was measured with the IA system connected to the OM. To ensure that both measurements were accurate, both devices were calibrated with standards. Calipers were calibrated with a National Institute for Standards and Technology (NIST)-traceable calibration gage block with a precision of 0.025 mm. The OM with IA was calibrated with an Olympus micrometric calibration ruler with the resolution of 0.01 mm. A calibration check was performed by measuring a series of pretest specimens with both methods. There was good agreement between the two methods (less than 3.5% relative difference for multiple measurements).

2.3. Data evaluation

After the test termination, specimens were removed from the vessel and divided into two parts with a diamond-impregnated saw. One part was used for OM/IA evaluation of the specimen cross-section, and the other part was used for phase identification with X-ray diffraction (XRD) and/or scanning electron microscopy (SEM) with energy dispersive spectroscopy (EDS). Since the remaining glass was translucent, transmitted light enables clear determination of the remaining glass from alteration layers (see Fig. 2). The remaining glass thickness was determined by performing at least 10 measurements equally distributed across the crack-free areas of the specimen. This step yields the average thickness of the remaining glass and the standard deviation of the measurements.

If the cross-section cut is not exactly perpendicular to the specimen surface, an increase in measured thickness will occur. A cut of 20° off perpendicular would increase the measured thickness by roughly 6%. The cut is not perpendicular if the thickness measured under transmitted light is different than that measured under reflected light. However, unless the cut is substantially different from 90°, the resulting error is negligible.

The mass of glass altered or converted into alteration products (m_a) per unit surface area is calculated with the following formula:

$$m_a = \frac{1}{2} d_i \rho \left(1 - \frac{d_r}{d_i}\right) = \frac{m_i}{2 w_i l_i} \left(1 - \frac{d_r}{d_i}\right), \qquad (2)$$

where d_i is the initial glass thickness, ρ is bulk density, d_r is the thickness of the remaining glass, w_i and l_i are the initial specimen width and length, respectively, and m_i is the initial specimen mass.

3. Results and discussion

3.1. General alteration curve

Given that a limited amount of water is condensed on the specimen surface during the VHT, the test can be viewed as a hydration of glass in aqueous solution at a high surface-area-to-solution volume ratio. Fig. 3 displays the general alteration curve for LAW glasses reacting with water in a closed system. The alteration process in a closed system can be divided into four distinguishable stages. In the first stage, glass reacts with the dilute solution at a roughly constant rate. This rate, the forward rate of reaction, is a function of temperature, solution chemistry (primarily pH), and glass composition. In a closed or slowly replenished system, the concentration of key reaction products increases in the solution as the reaction progresses. The second stage of reaction occurs when the concentration of key alteration products becomes high enough to significantly lower the reaction rate. The solution species most often credited for affecting the alteration rate is orthosilicic acid [15,16]. However, other species were observed to influence the alteration rate [8,17,18]. As the reaction progresses through the second stage of alteration, the concentration of

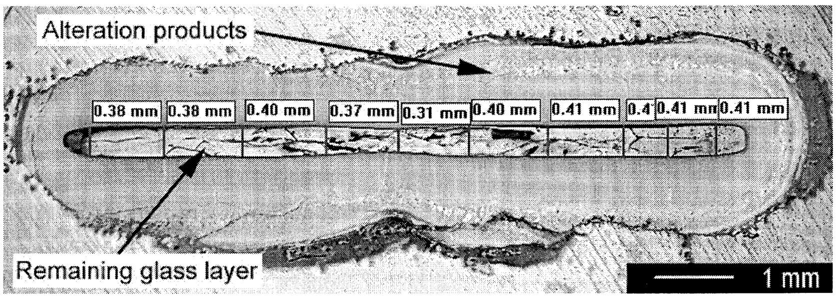

Fig. 2. OM/IA measurement of LAW-ABP1 after 2 days of VHT at 300 °C in combination of transmitted and reflected light (thickness of remaining glass is displayed above each measurement).

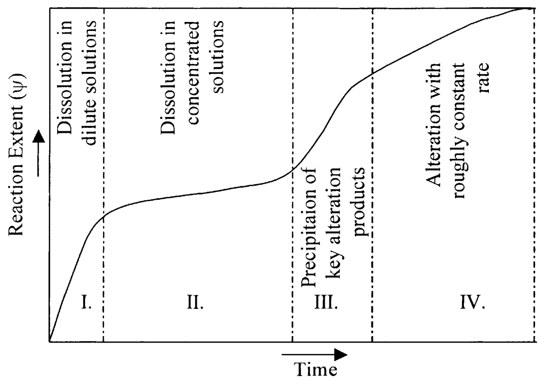

Fig. 3. General alteration curve.

key reaction products becomes supersaturated in solution with respect to a number of solid phases. At some point, one or more of these solid phases are nucleated and begin to precipitate. The growth of these phases from the supersaturated solution is generally fast. The solution concentrations of those components incorporated into solid alteration products decrease, thus increasing the driving force for dissolution, enhancing the alteration rate, and entering the third stage of alteration. The transition from the second to the third stage of alteration requires the nucleation and growth of certain key alteration products. These processes are influenced by a number of parameters that cannot be precisely controlled and therefore influence the reproducibility of the results. After this acceleration period, the fourth stage of alteration is encountered, and the process slows down to an approximately constant alteration rate. Tests conducted with different amounts of water resulted in similar alteration rates, indicating that the overall alteration rate may not be affected by the amount of water available for the reaction. However, there is typically a number of alteration layers generated on the sample during the tests, and therefore other processes, such as the transport of species through alteration layers, may influence the rate of alteration. The observed alteration rate was approximately constant during the fourth stage of alteration. The first three stages are widely recognized [19–21] and are described in detail in [22]. Evidence for the fourth stage is given in this paper.

3.2. Specimen preparation

A series of tests with different specimen-preparation techniques was performed to determine the effect of cracks, which were introduced into the specimens during the cutting, and the specimen surface finish on the VHT results. A diamond-impregnated saw was used to cut specimens of LD6-5412 glass with dimensions of $10 \times 10 \times 1.5$ mm^3. Water-lubricated SiC paper with different grit sizes and a polishing cloth with CeO_2 were used to obtain surface finishes ranging from fresh cut to 1 μm. All tests were performed in duplicate at 150 °C for 14 days. It is apparent from Fig. 4 that the fresh-cut surface represents a considerably higher surface area than the 1-μm surface finish and contains a significant number of cracks and imperfections. It suggests that the mass of glass altered will be higher because of higher surface area and low cutting speed and pressure. This, together with a better surface finish, will decrease the number of cracks present in the specimen after the test termination. The mass of glass altered, displayed in Fig. 5, was determined from remaining glass thicknesses measured on cross-sections of specimens using OM/IA.

Surprisingly, specimens with a surface finish from 240 grit to 600 grit showed almost the same mass of glass converted into alteration products, m_a, while specimens with a fresh-cut surface showed a smaller mass of glass altered, and specimens with a 1-μm surface finish did not react.

Examination of specimens with fresh-cut surfaces revealed a significant number of cracks (see Fig. 6(a)), and therefore, the alteration took place not only on the surface, but also inside the cracks. The measurement of the thickness of the remaining glass thus did not represent the actual mass of glass altered. A specimen with a 600-grit surface finish, displayed in Fig. 6(c), did not contain cracks. For specimens with a 1-μm surface finish, the incubation time, which is the time necessary to create suitable conditions for precipitation of alteration products, was longer than for specimens with a 600-grit surface finish. Therefore, the specimen did not form a measurable alteration-product layer within the test duration.

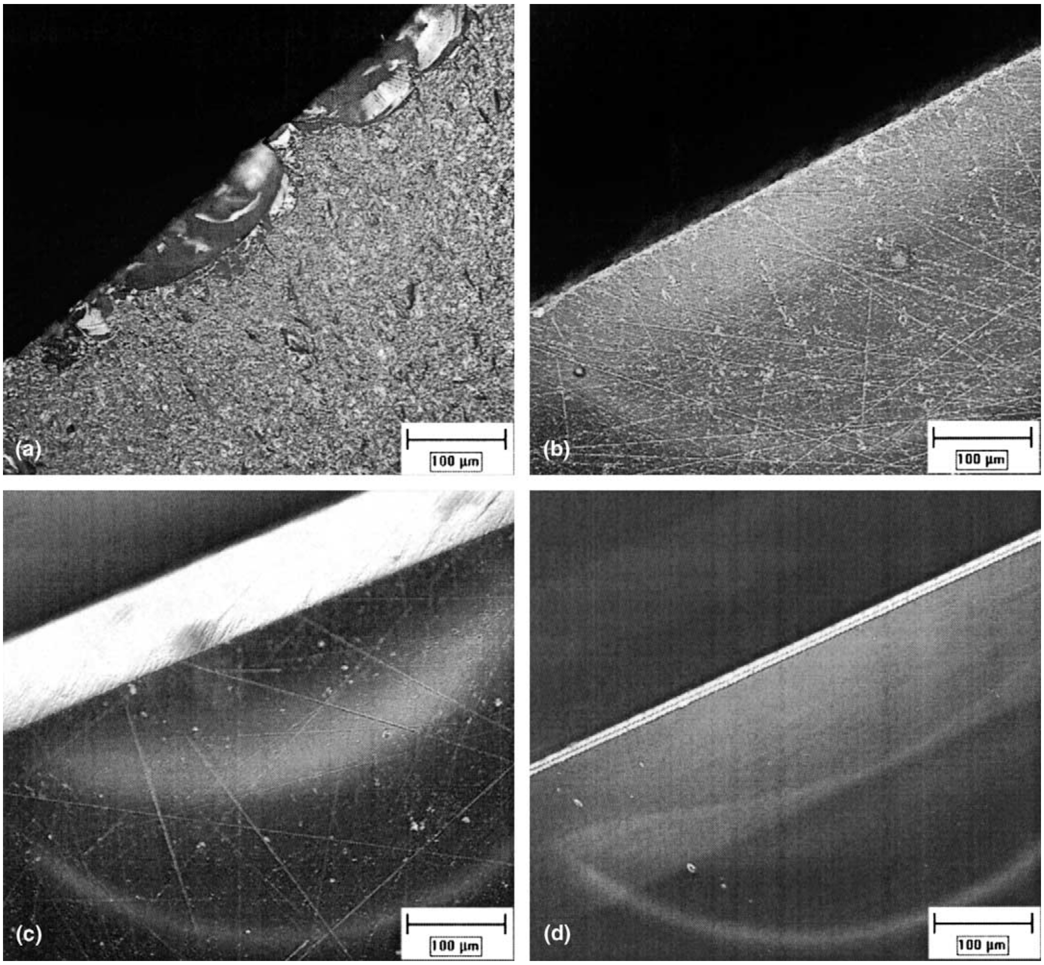

Fig. 4. Specimen surface: (a) Fresh cut with a diamond-impregnated saw; (b) 400-grit surface finish (SiC paper); (c) 600-grit surface finish (SiC paper); and (d) 1-μm surface finish (CeO_2).

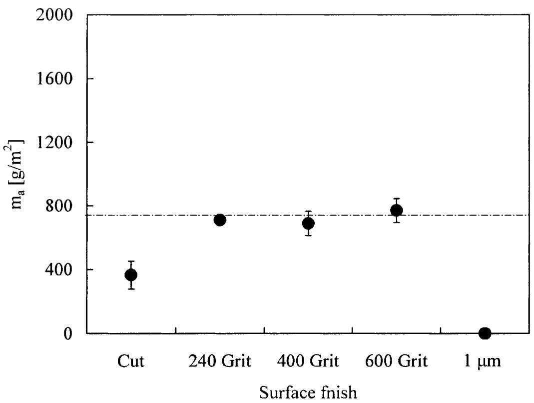

Fig. 5. LD6-5412, results of duplicate test with different surface finish at 150 °C, 14 days.

Based on the results presented above, it was decided to prepare all subsequent specimens for VHT with all sides polished to 600 grit. This procedure will decrease the number of imperfections/cracks present on the surface and enable comparison of the results with other authors, since 600 grit is a commonly used surface finish.

3.3. Specimen holder modification

The VHTs are typically conducted with specimens suspended with a Teflon or Pt wire. Teflon threads were found to break more easily during the test, thereby invalidating the test. Platinum wire is

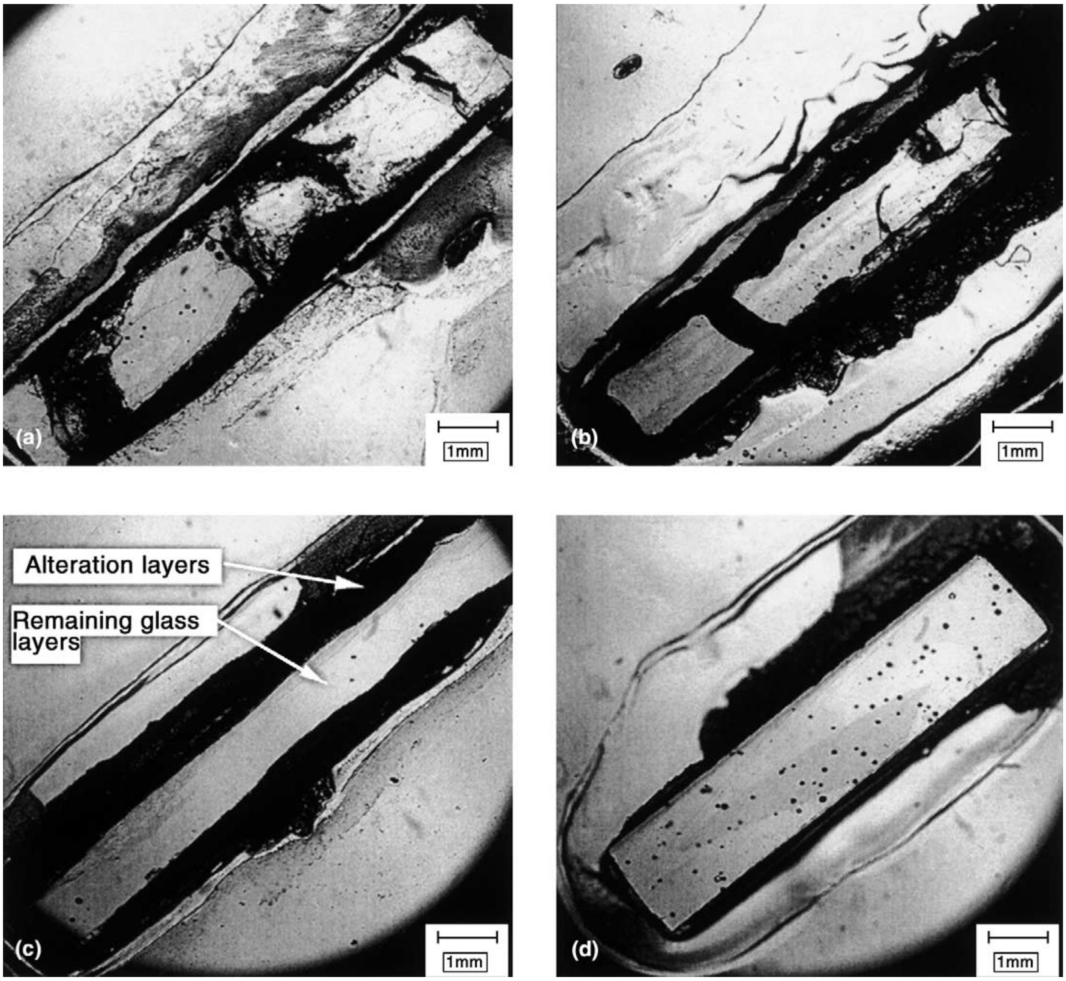

Fig. 6. Cross-sections of LD6-5412 specimens with a different surface finish after 14 days of VHT at 150 °C: (a) Fresh cut; (b) 240-grit surface finish; (c) 600-grit surface finish; (d) 1-μm surface finish.

more durable, but was observed to provide a nucleation center for the growth of alteration products. For this study, Pt wire was selected because of its resistance to thermal and mechanical stress and the ease of use.

The aim of tests with different specimen holders was to minimize the possibility of specimen dropping, minimize the area in contact with Pt wire, and possibly increase the reproducibility of the results. Four examples of tested specimen holders are displayed in Fig. 7.

When support notches were used (Fig. 7(a)), the specimen had more cracks after the test was terminated. Approximately 90% of notched specimens contained at least one crack. Significantly fewer specimens had cracks (about 10%) when specimen holders displayed in Fig. 7(b)–(d) were used. The holder with a minimum surface area in contact with Pt wire (Fig. 7(b)) and the holder with rounded edges (Fig. 7(c)) did not show any significant improvements in test reproducibility/results when compared to the holder displayed in Fig. 7(d). Since this holder is also the easiest to use, it was selected as the most suitable setup for conducting the VHT.

The number of tests with elevated pH, detected after the test was terminated, decreased roughly

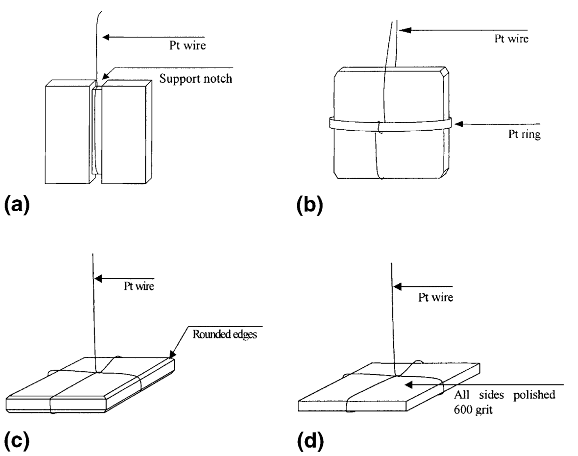

Fig. 7. Examples of tested specimen holders.

by 50% when specimens were positioned in the horizontal configuration (Fig. 7(d)), compared to those positioned in the vertical configuration. Elevated pH was used as an indicator of possible reflux during the test and/or contamination of the condensate water from spallation of alteration products off the specimen. Reflux during the test would enable transport of soluble species from the specimen surface to the inside of the vessel and modify the solution chemistry on the specimen surface. Therefore, the volume of water needed for conducting these tests was determined experimentally, so no reflux/mass exchange would be likely to occur. A certain volume of water condenses on the specimen during cooling, and the horizontal position of the specimen can reduce the probability of dripping during the termination, which would also increase the pH of residual liquid.

3.4. Incubation time and rate evaluation

The incubation time is the time needed for the vapor-hydration process to establish suitable conditions for solid alteration products to precipitate. Its existence also plays an important role in the test reproducibility, which will be discussed later. During the incubation time, the first two stages of alteration take place (see Fig. 3). The alteration enters the third stage with precipitation of key alteration products. The alteration products start to form on specimen surfaces from nucleation centers and extend across the specimen, eventually forming a continuous layer, typically 50–300-μm thick. A characteristic alteration curve of LAW glasses subjected to VHT is displayed in Fig. 8. The shape of the alteration curve resembles the third and fourth stage of alteration displayed in Fig. 3. The alteration process in these stages consists of (1) the incubation time, (2) precipitation of first alteration products indicated by an increase in alteration rate, and (3) a linear $m_a - t$ region with a roughly constant rate.

Specimens of LD6-5412 entering the third stage of alteration are displayed in Fig. 9. Specimen (a) shows a thin 30-μm layer, where the alteration products start to precipitate (white dots). Specimen (b) shows an almost continuous alteration layer roughly 100 μm thick.

A specimen of LD6-5412 glass after 3 days of VHT at 175 °C is displayed in Fig. 10. Although the test period was only 24 h longer than in Fig. 9, the specimen shows a continuous layer of alteration products about 600 μm thick (see cross-section in Fig. 10(b)). An acceleration of the process occurred between the second and third day of the test, resulting in the formation of a uniform thick layer of alteration products on the surface of the specimen. Before the acceleration took place, the nucleation and formation of key alteration product(s) had to occur. It is not exactly clear what processes affect the time needed to precipitate the

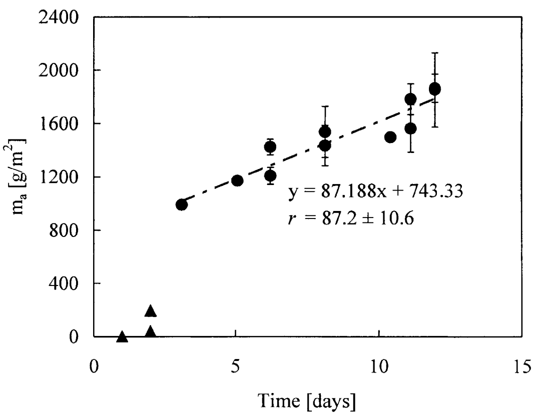

Fig. 8. Results from VHT on LD6-5412 glass at 175 °C.

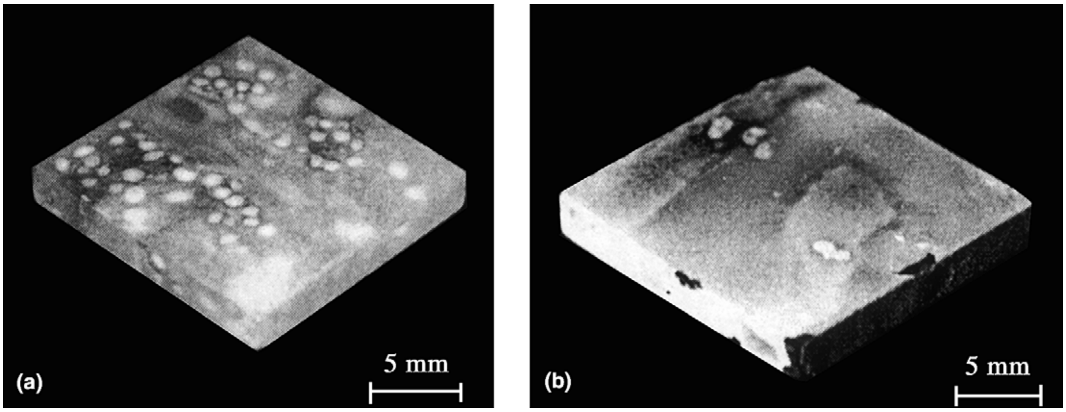

Fig. 9. Duplicate specimens of LD6-5412 glass after 2 days of VHT at 175 °C.

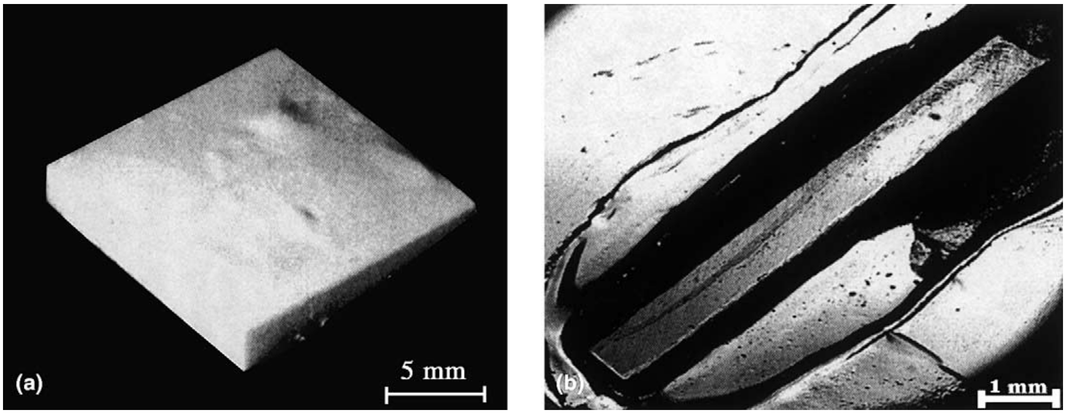

Fig. 10. (a) Specimen of LD6-5412 glass after 3 days of VHT at 175 °C. (b) Cross-section.

key alteration product, but the comparison of duplicate specimens (Fig. 9) in the acceleration region suggests that there is a significant scatter in the data.

Sudden increases in alteration rate were observed, for example, by Abrajano et al. [23], who used VHT to study the corrosion of nuclear waste glasses. Since their study was conducted mainly on specimens approaching the third stage of alteration (the thickness of alteration layers was approaching 50 μm), none of the specimens entered the fourth stage of alteration. Those specimens were excluded from evaluation by Abrajano et al. [23] because of the sudden increase in the alteration rate and were subsequently marked as anomalous. However, this observation corresponds to the third stage of alteration as displayed in Fig. 8 and confirms its existence.

The alteration rate increases after the key alteration products begin to precipitate on the surface of the specimen and becomes approximately constant after the alteration enters the fourth stage. The alteration rate was determined by regression over the linear portion of the alteration curve from 3 to 12 days (see Fig. 8). This alteration rate includes ion exchange processes, glass-matrix dissolution, and precipitation of alteration products. All of these reactions occur in a solution of evolving chemistry, which precludes isolation of the rate-limiting step. The term ‘rate of alteration’ is used within this paper as a rate at which glass is converted into the alteration products.

3.5. Volume of water

Selecting the appropriate water volume is of fundamental importance for establishing suitable and reproducible conditions in the VHT. If the relative humidity inside the vessel is too low, little reaction will occur. Abrajano et al. [23] observed a negligible reaction below 70% relative humidity (RH) at 202 °C. If the volume of water inside the vessel is too high, a reflux can occur, allowing the transport of soluble species from the surface of the specimen to the solution at the bottom of the vessel. Our aim was to calculate and experimentally verify the volume of water needed to conduct the tests at temperatures up to 300 °C at a maximum alteration rate with no reflux. Tests up to 175 °C were conducted with the volume of water calculated from steam tables [24] plus 0.05 ml of excess water per specimen. The pH of the residual liquid in the vessel after the test was selected as a measure of reflux or dripping from the specimen. The pH was determined with a pH paper because of the small volume of residual liquid. Since the tests up to 175 °C did not show elevated pH, or any other indication of possible reflux, tests with a different volume of water were conducted at 200, 250, and 300 °C. The volume of water, $\Delta V(H_2O)$, represents the volume difference from 100% RH calculated from steam tables.

Results from duplicate tests on LAW-A33 glass at 200 °C are displayed in Fig. 11. The volume of water used in VHT varied from −0.04 ml below 100% RH to 0.12 ml above RH. The mass of glass altered shows a slight increase at 100% RH and elevated pH for all tests conducted with higher volumes of excess water. Zero ml excess of water was selected as the most suitable for conducting VHT at 200 °C.

The VHTs on LAW-A33 at 250 °C were conducted with volumes of excess water from −0.20 to 0.10 ml. Results displayed in Fig. 12 indicate that the maximum amount of glass was altered in the specimen with −0.10 ml of excess water. Tests conducted with a volume of excess water greater

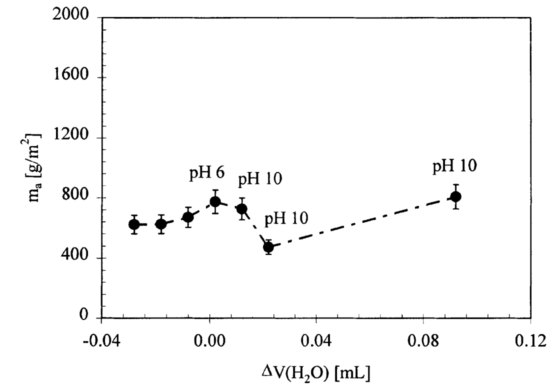

Fig. 11. The effect of excess water on the amount of LAW-A33 glass reacted in 7 days at 200 °C.

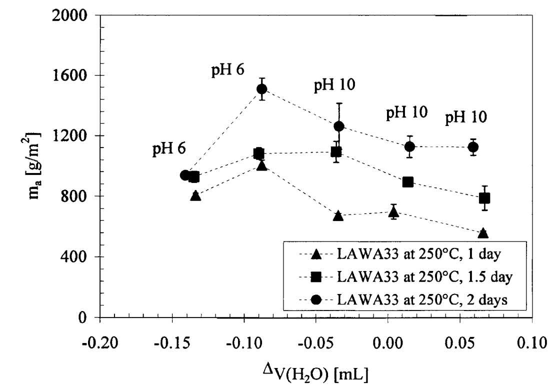

Fig. 12. The effect of excess water on the amount of LAW-A33 glass reacted in 1, 1.5, and 2 days at 250 °C.

than −0.10 ml showed elevated pH and contamination of the vessel with alteration products.

The alteration rates, calculated from data in Fig. 12, are plotted in Fig. 13. The alteration rates are similar, but the intercept decreases, or the time to reach the same reaction extent increases, with an increasing volume of excess water. The intercept ranges from 18 to 430 g/m^2, and the difference in time to achieve the same reaction extent is approximately 1 day.

The VHTs on LAW-ABP1 at 300 °C were conducted with volumes of excess water from −0.80 to 0.20 ml. Results displayed in Fig. 14 indicate that the highest alteration rate occurs roughly at −0.65 ml of excess water. However, −0.54 ml was selected as the most suitable volume

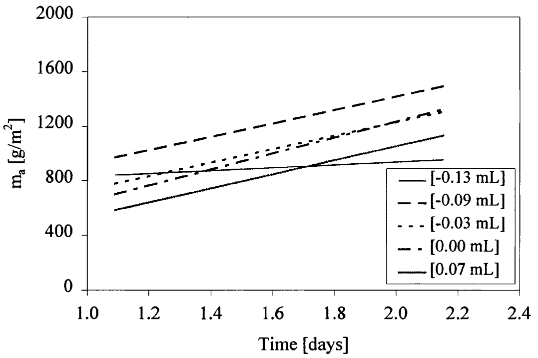

Fig. 13. Alteration of LAW-A33 at 250 °C for different volumes of excess water.

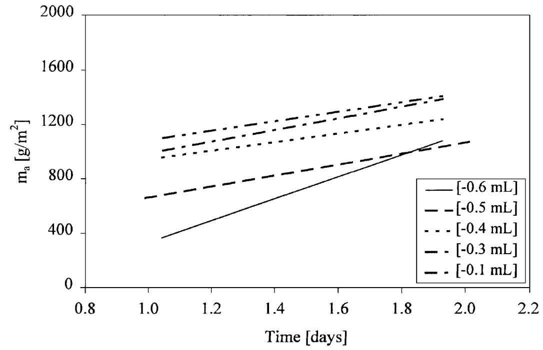

Fig. 15. Alteration of LAW-ABP1 at 300 °C with different volumes of excess water.

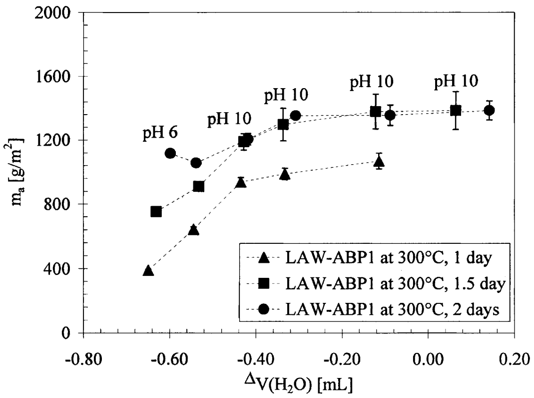

Fig. 14. The effect of excess water on the amount of LAW-ABP1 glass reacted in 1, 1.5, and 2 days at 300 °C.

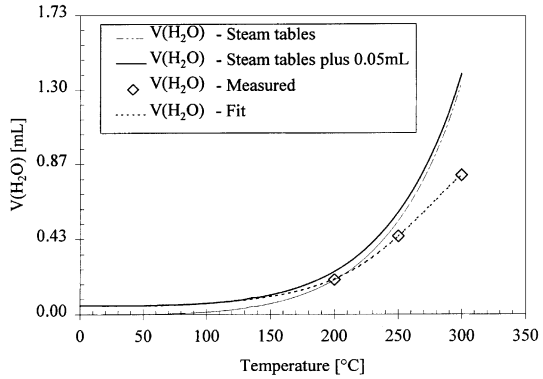

Fig. 16. Volume of water calculated from steam tables (100% RH plus 0.05 ml of excess water) compared to the volume of water measured experimentally.

of water for conducting the test because it is the highest volume of water for which there was no pH increase in the condensed water at the end of the test.

Similar to the tests at 250 °C, the alteration rates are comparable for all volumes of excess water. Surprisingly, the intercept increases with an increasing volume of excess water as shown in Fig. 15. The intercept ranges from 255 to 736 g/m^2, and the difference in time to achieve the same reaction extent is approximately 2 days. This trend is opposite to what was observed at 250 °C (see Fig. 13).

The volume of water calculated from steam tables to achieve 100% RH plus 0.05 ml of excess water and measured values fitted by empirical function are displayed in Fig. 16. As shown in Fig. 17, the volume of water required to perform the test differs from the 100% RH in the range from +0.05 ml for $T \leqslant 150$ °C to 0.54 ml for 300 °C. As was shown previously, the volume of water used to conduct the test does not have a significant impact on the rate itself, but affects the intercept and time that is needed to achieve the same reaction extent. Fig. 18 shows the determined volumes of water for 250 and 300 °C.

3.6. Volume of water effect

The VHT response of LAW-A33 at 200 °C was measured with two specimens per vessel and one specimen per vessel. One specimen of LAW-A33

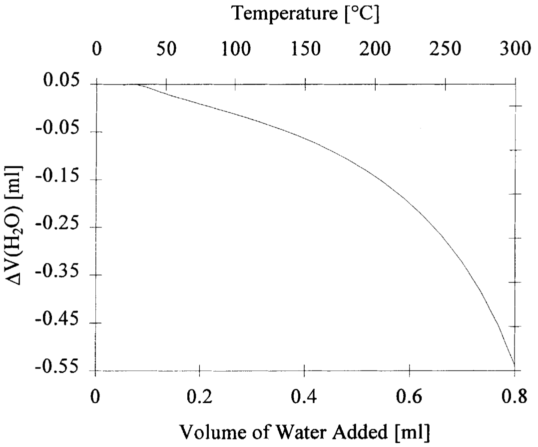

Fig. 17. Difference in volume of water calculated from experimental data and volume of water given in steam tables for achieving 100% RH, plotted vs. temperature.

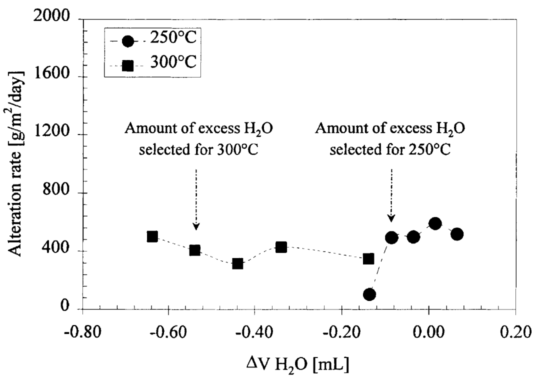

Fig. 18. Volume of excess water selected for conducting the VHT at 250 and 300 °C.

and one specimen of LAW-ABP1 were tested together in each vessel along with 0.25 ml of water (0.05 ml of water was added for each specimen to the 0.15 ml calculated for achieving 100% RH). The rate of LAW-ABP1 is roughly $10\times$ lower than that of LAW-A33. After roughly 25 days, LAW-A33 was completely reacted while LAW-ABP1 was still within its incubation period. The excess water available for corrosion of LAW-A33 was twice as high due to the low rate of LAW-ABP1.

Tests with one specimen of LAW-A33 per vessel were conducted with 0.20 mL of water. Fig. 19

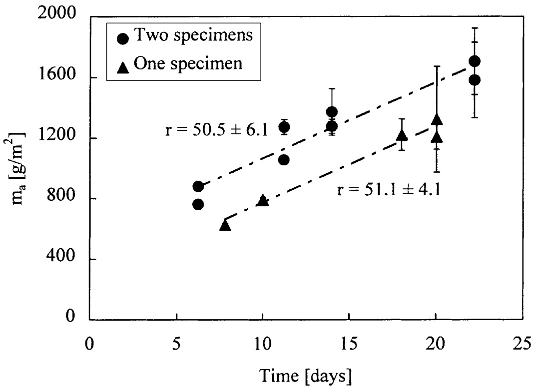

Fig. 19. The m_a as a function of t for LAW-A33 glass, tested at 200 °C with one and two specimens per vessel.

compares the m_a for those two sets of tests. The m_a values in tests with two specimens per vessel are clearly higher than those measured in tests with one specimen per vessel. However, the rates are the same. The difference in m_a is attributed to the difference in available water to corrode the LAW-A33 specimens. In this case, the intercept increased with an increasing volume of added water, and the difference between the times needed to achieve the same reaction extent was approximately 7 days.

3.7. Remaining glass thickness vs. alteration layer thickness

To determine the amount of glass converted into alteration products, it is possible to measure the remaining glass thickness or the thickness of alteration layers. LAW-A44 glass was measured because it follows the typical behavior of LAW glasses subjected to VHT conditions, and the alteration layers are easy to distinguish.

Cross-sections of LAW-A44 glass subjected to VHT at 250 °C for different test durations are displayed in Fig. 20. It is apparent that the thickness of remaining glass (in the middle) decreases with increasing reaction time, while the thickness and number of alteration layers increases with time.

The remaining glass thickness, the alteration-layer thickness, and the whole specimen thickness were measured and plotted versus time in Fig. 21.

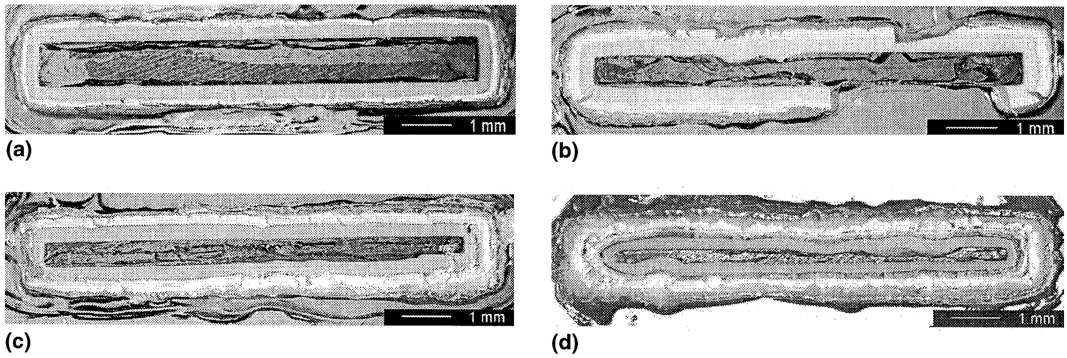

Fig. 20. Cross-sections of LAW-A44 specimens after the VHT termination: (a) LAW-A44 after 5 days of VHT at 250 °C; (b) LAW-A44 after 8 days of VHT at 250 °C; (c) LAW-A44 after 10 days of VHT at 250 °C; (d) LAW-A44 after 14 days of VHT at 250 °C.

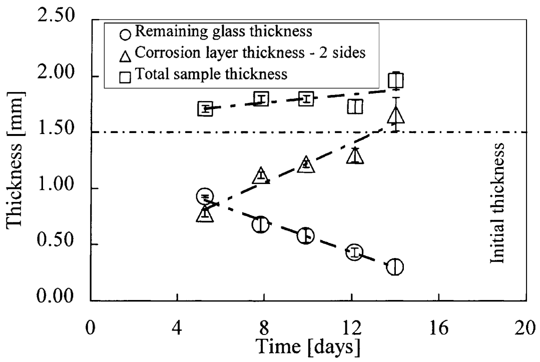

Fig. 21. Comparison of remaining glass thickness and corrosion layer thickness.

The results show that the thickness of the whole specimen increases with reaction progress because of the difference in bulk density of the glass and alteration products. In this case, the specimen, after 14 days of VHT at 250 °C, is almost 2 mm thick, which is equal to a 25% increase in thickness compared to the initial 1.5 mm. Although an increase in thickness is usually observed, its extent depends on the bulk density of alteration products. This significantly contributes to the error connected to the measurement of alteration-layer thickness and complicates comparisons between different glasses based just on alteration-layer measurement.

The bulk density of alteration products was estimated from IA by measuring the difference in the specimen's size before and after the test and assuming no mass change of the specimen. The bulk density of the alteration layers was found to range from 1.08 g/cm^3 for LD6-5412 to 1.34 g/cm^3 for LAW-A44, while the densities of initial glasses were 2.54 g/cm^3 for LD6-5412 and 2.68 g/cm^3 for LAW-A44. Because it is known that the chemistry, morphology, density, and general characteristics of alteration layers are a strong function of the chemistry of the glass and leaching conditions [25], the alteration-layer thickness cannot be used to compare the reaction extents of different glasses. If two different glasses show a different thickness of alteration layers after the same test period, it cannot be concluded that one glass altered more or less than the other because the thickness of alteration layers cannot be directly converted to the amount of glass altered.

The alteration layers are generally not uniform (see Fig. 22), which increases the uncertainty of the thickness measurement. The interface between the initial glass and the alteration layers is more uniform than the surface of the alteration layers.

In contrast, the measurement of the remaining glass-layer thickness is directly related to the amount of glass converted into the alteration products. However, a significant drawback is that more extensive glass alteration is required before sufficient changes in the remaining glass thickness occur to provide reasonable corrosion-rate estimates. Also, the technique is not suitable for glasses experiencing an unusual amount of cracking during the test.

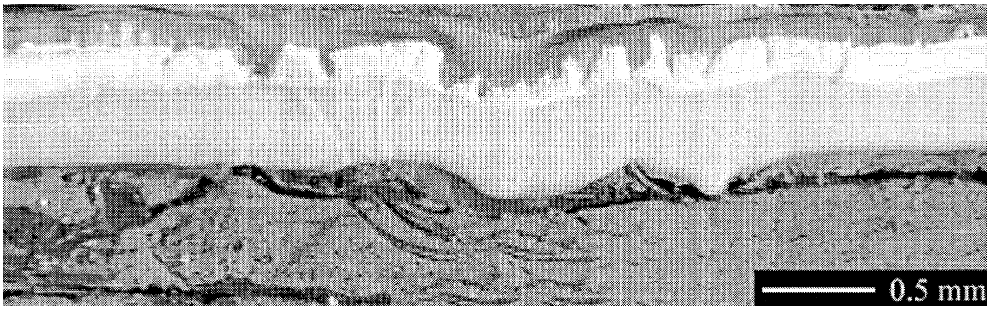

Fig. 22. Detail of the interface between glass and alteration layers, LAW-A44 after 8 days of VHT at 250 °C.

3.8. Measurement precision

The OM with IA was calibrated with an Olympus micrometric calibration ruler with a resolution of 0.01 mm. An SEM can be calibrated to a resolution that is at least an order of magnitude better than the IA. In the third stage of alteration when crystalline phases start to precipitate on the surface of the specimen, the alteration layers cannot be measured with IA because the system cannot be calibrated with sufficient precision. In this case, an SEM is a more suitable tool for measurement and characterization of alteration layers.

After the alteration products form a continuous layer and the alteration process enters the fourth stage of alteration, the thickness of the alteration layers is typically greater than 300 μm, and therefore, the precision of the IA system is sufficient for performing the measurements.

3.9. Comparison with published results

To compare data measured according to this procedure to results reported in the literature [12], the differences between experimental methods have to be identified. Table 2 summarizes the test parameters for both methods. Two differences between the methods include the manner in which the alteration extent is evaluated and the specimen dimensions.

To directly compare the results measured according to both procedures, the results reported in the literature had to be recalculated to yield the mass of glass altered in g/m^2. Because the thickness of the specimen before the test was 1000 μm,

Table 2
Comparison of test parameters

	Reported in [10,12]	This study
Specimen dimensions	Core drilled specimen Diameter = 10 mm Thickness = 1 mm	Square specimen Width = 10 mm Length = 10 mm Thickness = 1.5 mm
Surface finish	One side polished to 600 grit	All sides polished to 600 grit
Mass of glass (g)	0.20	0.39
m_a = mass of glass per unit surface area (g/m^2)	1300	1950
Volume of water added to 22 ml vessel with two specimens at		
T = 150 °C	0.15	0.15
T = 175 °C	0.20	0.19
T = 200 °C	0.25	0.25
Evaluation method	Thickness of alteration layer	Thickness of remaining glass

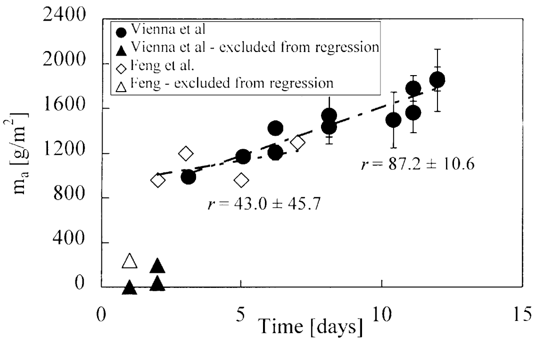

Fig. 23. Comparison of VHT results on LD6-5412 at 175 °C from literature [10,11] and this study [14].

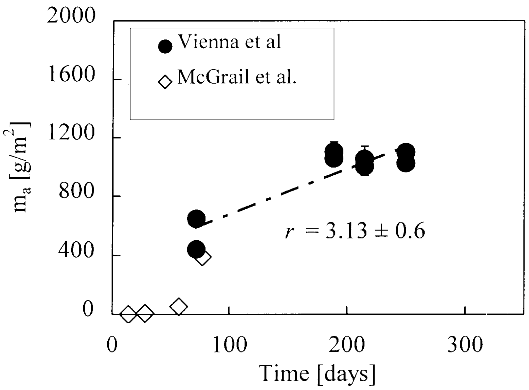

Fig. 24. Comparison of VHT results on LAW-A33 glass at 150 °C from [13,14].

and the maximum alteration extent was 1300 g/m^2, we can assume that 0 μm corresponds to 0 g/m^2 of altered glass, a 1000 μm-thick alteration layer corresponds to 1300 g/m^2 of altered glass, and they are linearly related. This approximation enables direct comparison of the results.

Fig. 23 compares results from [12,14] for LD6-5412 subjected to the VHT at 175 °C. The tests were performed with two specimens per vessel and 0.19 ml of water in [12] (0.05 ml of water was added for each specimen to the 0.9 ml calculated for achieving 100% RH) and with one specimen per vessel and 0.14 ml of water in [14]. Qualitatively, there is good agreement between the m_a values. However, the calculated rate for the data found in [12], 42.9 g/m^2/d, is significantly lower than that of the 87.3 g/m^2/d calculated from [14].

Fig. 24 compares results for LAW-A33 glass at 150 °C reported in [14] to the results reported in [13]. Tests in [13] were performed with two specimens per vessel and 0.15 ml of water (0.05 ml of water was added for each specimen to the 0.05 ml calculated for achieving 100% RH). The tests in [14] were performed with one specimen per vessel and 0.10 ml of water. The mass of glass altered is in very good agreement and confirms the existence of the incubation time ending at roughly 70 days. Insufficient data were reported in [13] to calculate the alteration rate.

3.10. Comparison of VHT and PCT

From the parameters that can be obtained from VHT, the time it takes for the glass to enter the third stage of alteration appears to be one of the parameters that might be successfully compared with other methods. Table 3 shows the comparison of acceleration times between VHT and PCT. The VHTs were conducted with one specimen per vessel at 200 °C [14]. The PCTs were conducted with deionized water at 99 °C with $S/V = 20\,000$ m^{-1} [13].

The time it takes to reach the third stage of alteration, plotted in Fig. 25, was obtained from VHT as the time when the first solid alteration products were observed. In PCT, this is the time where the normalized mass loss (calculated from boron release) increases rapidly on a log-scale plot.

Table 3
Comparison of the estimated VHT and PCT acceleration times

Glass ID	Temperature (°C)	VHT acceleration time (days)	PCT acceleration time (days)
LD6-5412	200	$\leqslant 1$	13
LAW-A33	200	$\leqslant 7$	>63
HLP-47	200	$\leqslant 5$	>41
LAW-ABP1	200	$\leqslant 6$	>174

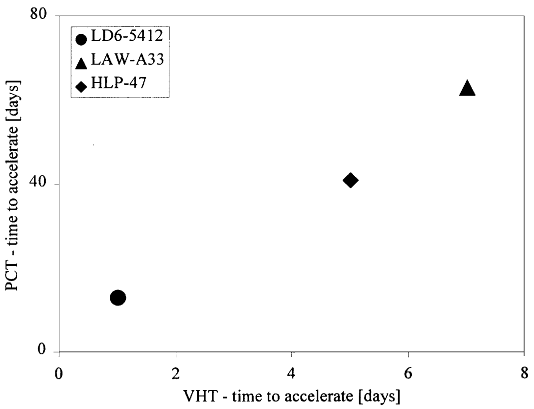

Fig. 25. Time to accelerate comparison.

However, the resulting values are only estimated because neither of the tests are designed to precisely determine the time it takes to accelerate. In PCT, some glasses may take a long time to accelerate or possibly not accelerate at all. The HLP-47 did not accelerate before 41 days, but dissolved completely after 150 days. The LAW-A33 did not accelerate before 63 days, but dissolved completely after 170 days. The time to accelerate cannot be determined exactly because of insufficient experimental results. LAW-ABP1 is not included in the plot because the alteration process did not accelerate even after 277 days, and therefore, it might be one of the glasses that does not accelerate in PCT.

Although the data show a limited amount of correlation, the existence of the third stage was observed in PCT, VHT, and other tests used for testing LAW glasses. This may provide a tool that could be used in comparing these tests.

3.11. Reproducibility

Four glasses with the same target composition were fabricated for testing the reproducibility of the measurement within our lab. Their composition is given in Table 1 as HLP-01. Glasses were tested at 200, 250, and 300 °C. Measured alteration rates are given in Table 4.

Comparison of the data at 200 °C for the baseline glass is plotted in Fig. 26. The rates do not show a good agreement and range from 0.2 to 5.7 g/m^2/d. It is apparent that although the rates show an order of magnitude difference, the m_a values are comparable. In other words, comparing m_a values for the four glasses suggests that they behave similarly; however, the fitted rates vary between glasses. The main reasons for the differences in rate are the number and position of measured data points. Glass HLP-43 exhibits the lowest alteration rate as only four points are available, and all of them are close to the 400 g/m^2 of glass altered, resulting in a rate of 0.2 g/m^2/d. This glass shows a high resistance to VHT at 200 °C, so it was not possible to dissolve more than 800 g/m^2 before significant loss of water in the test vessel occurred.

The alteration rates measured at 250 °C for the baseline glass are compared in Table 4 and plotted in Fig. 27. The lowest alteration rate, measured for HLP-26, is 224 g/m^2/d. The highest alteration rate, measured for HLP-25, is 292 g/m^2/d. The average alteration rate was calculated as 254 ± 35.4 g/m^2/d. The percent relative standard deviation (% RSD) for the thickness variability of a single specimen, the 10 measurements performed on each specimen, is 2.9%. The variability in the

Table 4
Comparison of alteration rates measured at 200, 250, and 300 °C for the baseline glass

Glass ID	Alteration rate at 200 °C (g/m^2/d)	Alteration rate at 250 °C (g/m^2/d)	Alteration rate at 300 °C (g/m^2/d)
HLP-01	2.4 ± 0.4	264 ± 18.7	505 ± 80.0
HLP-25	5.7	29222.9	655 ± 166
HLP-26	1.3 ± 0.4	224 ± 13.8	686 ± 94.5
HLP-43	0.2 ± 0.9	234 ± 7.5	721 ± 146
Combined	2.8 ± 0.4	254 ± 35.4	642 ± 94.9
%RSD[a]	99.0	14.6	14.8

[a] Percent relative standard deviation accounts for differences in multiple measurements of each specimen, differences in points from the $m_a - t$ line, and differences between glasses.

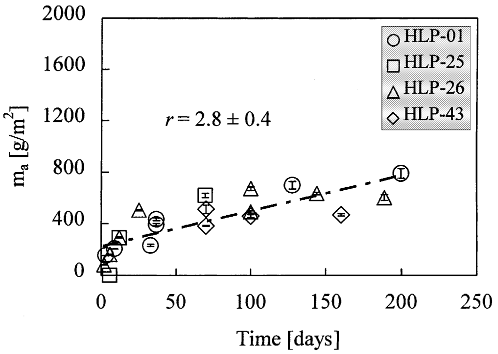

Fig. 26. Comparison of the measured data at 200 °C.

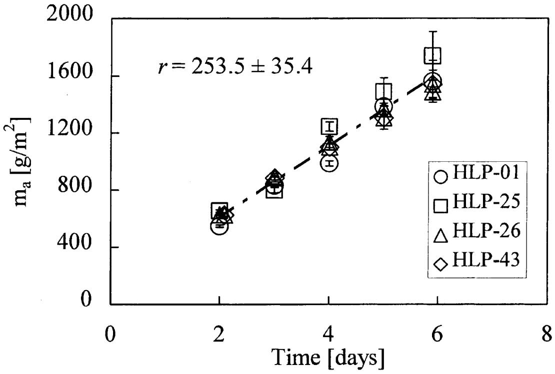

Fig. 27. Comparison of VHT results for baseline glasses at 250 °C.

measured m_a values over the 2 months is 7.7% RSD. The variability of the rate over the 2 months for these tested glasses is 14.6% RSD.

The alteration rates measured at 300 °C for the baseline glass are compared in Table 4 and plotted in Fig. 28. The average alteration rate was calculated as 642 ± 94.9 g/m^2/d.

As discussed above, the lowest testing temperature, 200 °C, shows the highest uncertainty in alteration rate. This is caused mainly by the relatively high resistance of the baseline glass to VHT at 200 °C where the test vessel loses a significant amount of water during the test before the m_a value of 800 g/m^2 can be reacted. The alteration rate is also influenced by the number and distribution of the measured data points, which further increases the uncertainty. Higher temperatures

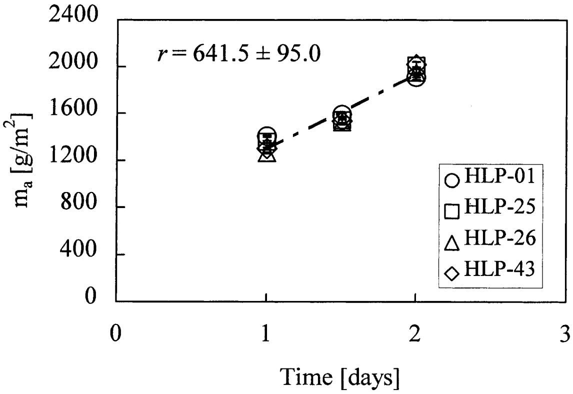

Fig. 28. Comparison of VHT results for baseline glasses at 300 °C.

enable the specimens to fully react before a significant amount of water is lost, thus yielding more consistent and reproducible results.

4. Conclusion

In this study, VHTs on LAW glasses were conducted at different temperatures with different amounts of water, surface finishes, and specimen holders. Both the thickness of alteration layers and the thickness of the remaining glass were used to evaluate the results with IA and SEM. Results from this study were compared to the results published in the literature. The reproducibility of the test was assessed by conducting a series of tests on four LAW glasses with the same target composition.

The results indicate that the VHT can be conducted with 100% relative humidity and 0.5 ml of excess water per specimen up to 150 °C. Above this temperature, the volume of added water must be determined experimentally. However, all tests in this study were conducted on LAW glasses, and it is possible that these values will change, depending on the materials that are tested. It is, therefore, recommended to determine the correct amount of water for specific materials. The volume of added water affected the mass of glass altered, but did not seem to affect the alteration rate in the fourth stage of alteration.

The alteration process of LAW glasses can be generally divided into four stages. The VHT is focused on measurements between the third and fourth stage and could be possibly used to determine the time that is needed to reach the third stage of alteration. This would enable the comparison of the results with other durability tests.

Data from VHTs presented in this paper were shown to be consistent with previously published results. However, the evaluation method and the representation of measured data were simplified and improved by use of the IA system. Specimens with all sides polished to a 600-grit surface finish are suitable for conducting the test and enable direct comparison of alteration rates for different glasses.

The mass of glass altered/converted to alteration products, per unit surface area, can be determined from the specimen cross-section by measuring the thickness of the remaining glass. The alteration rate can be determined from the linear portion of the alteration curve ($m_a - t$ dependence) and represents the rate at which glass is converted into the alteration products.

Acknowledgements

The authors would like to acknowledge the help of W.L. Ebert for helpful discussions and insights that helped guide this work. Ian Peg provided helpful discussions and compositions of LAW-A33 and LAW-A44 glass compositions. A.J. is grateful to Associated Western Universities for the fellowship appointment at Pacific Northwest National Laboratory. This study was funded by the US Department of Energy's Offices of Science and Technology through the Tanks Focus Area and River Protection. Pacific Northwest National Laboratory is operated for the US Department of Energy by Battelle under Contract DE-AC06-76RL01830.

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